0.1 Grundfunktionen im rechtwinkligen Dreieck

Definitionen (G = Gegenkathete, A = Ankathete, H = Hypotenuse)

$$\sin(\alpha) = \frac{G}{H} \qquad \cos(\alpha) = \frac{A}{H} \qquad \tan(\alpha) = \frac{G}{A} = \frac{\sin(\alpha)}{\cos(\alpha)} \qquad \cot(\alpha) = \frac{A}{G} = \frac{\cos(\alpha)}{\sin(\alpha)} = \frac{1}{\tan(\alpha)}$$

0.2 Sinus- und Kosinussatz

Sinussatz:

$$\frac{a}{\sin(\alpha)} = \frac{b}{\sin(\beta)} = \frac{c}{\sin(\gamma)}$$

Kosinussatz:

$$a^2 = b^2 + c^2 - 2bc\cos(\alpha) \qquad b^2 = a^2 + c^2 - 2ac\cos(\beta) \qquad c^2 = a^2 + b^2 - 2ab\cos(\gamma)$$

SSW-Fall (Sinussatz): Es können zwei Lösungen auftreten, da $\sin(\beta) = \sin(180° - \beta)$.

0.3 Trigonometrische Identitäten

0.3.1 Beziehungen zwischen Sinus und Kosinus

$$\sin(x) = \cos\left(x - \tfrac{\pi}{2}\right) \qquad \cos(x) = \sin\left(x + \tfrac{\pi}{2}\right) \qquad -\sin(x) = \cos\left(x + \tfrac{\pi}{2}\right) \qquad -\cos(x) = \sin\left(x - \tfrac{\pi}{2}\right)$$

0.3.2 Pythagoräische Identität und Symmetrie

$$\sin^2(\alpha) + \cos^2(\alpha) = 1 \qquad \cos(x) = \cos(-x) \qquad \sin(x) = -\sin(-x)$$

0.3.3 Additionstheoreme

Additionstheoreme

Sinus:

$$\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \sin(\beta)\cos(\alpha)$$

Kosinus:

$$\cos(\alpha + \beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta) \qquad \cos(\alpha - \beta) = \cos(\alpha)\cos(\beta) + \sin(\alpha)\sin(\beta)$$

Tangens:

$$\tan(\alpha + \beta) = \frac{\tan(\alpha) + \tan(\beta)}{1 - \tan(\alpha)\tan(\beta)} \qquad \tan(\alpha - \beta) = \frac{\tan(\alpha) - \tan(\beta)}{1 + \tan(\alpha)\tan(\beta)}$$

Kotangens:

$$\cot(\alpha + \beta) = \frac{\cot(\alpha)\cot(\beta) - 1}{\cot(\beta) + \cot(\alpha)} \qquad \cot(\alpha - \beta) = \frac{\cot(\alpha)\cot(\beta) + 1}{\cot(\beta) - \cot(\alpha)}$$

0.3.4 Doppelwinkeltheoreme

Spezialfall der Additionstheoreme für $\alpha = \beta$:

$$\sin(2\alpha) = 2\sin(\alpha)\cos(\alpha)$$

$$\cos(2\alpha) = \cos^2(\alpha) - \sin^2(\alpha) = 2\cos^2(\alpha) - 1 = 1 - 2\sin^2(\alpha)$$

$$\tan(2\alpha) = \frac{2\tan(\alpha)}{1 - \tan^2(\alpha)} \qquad \cot(2\alpha) = \frac{\cot^2(\alpha) - 1}{2\cot(\alpha)}$$

0.3.5 Potenzen — nützlich für Integration

Quadratische Ausdrücke (aus Doppelwinkeltheorem)

$$\sin^2(\alpha) = \tfrac{1}{2}\big(1 - \cos(2\alpha)\big) \qquad \cos^2(\alpha) = \tfrac{1}{2}\big(1 + \cos(2\alpha)\big)$$

0.3.6 Produktformeln

$$\sin x \cos y = \tfrac{1}{2}\big(\sin(x+y) + \sin(x-y)\big)$$

$$\cos x \cos y = \tfrac{1}{2}\big(\cos(x+y) + \cos(x-y)\big)$$

$$\sin x \sin y = \tfrac{1}{2}\big(\cos(x-y) - \cos(x+y)\big)$$